

Name: _____

Period: _____

Date: _____

Topic 1.1 Guided Notes: Interaction of Heredity and Environment

Unit 1: Biological Bases of Behavior | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appspsychreviewhub.com/overviews/topic1.1

Word Bank

<i>Heredity</i>	<i>Heritability</i>	<i>Dizygotic Twins</i>	<i>Genetic Predisposition</i>
<i>Epigenetics</i>	<i>Monozygotic Twins</i>	<i>Twin Studies</i>	<i>Natural Selection</i>
<i>DNA</i>	<i>Evolutionary Perspective</i>	<i>Family Studies</i>	<i>Eugenics</i>
<i>Nurture</i>	<i>Adoption Studies</i>	<i>Nature</i>	<i>Population</i>

1. The Dynamic Duo: Nature and Nurture

- _____: the influence of our inherited characteristics (genetics) on personality, physical growth, and social interactions — our biological blueprint.
- _____: the actual mechanism at work — the genetic transfer of characteristics from parents to offspring.
- _____ (Deoxyribonucleic Acid): the molecule that carries the genetic instructions for the development, functioning, and inheritance of traits.

Genetic instructions don't guarantee behavior — instead they provide a _____, an increased likelihood of developing a particular trait based on a person's genetic makeup (e.g. inheriting the potential for musical ability, which still needs the right environment to develop).

- _____: the influence of environmental factors such as family, culture, and experience on development.

The Bridge: Epigenetics

_____ is the study of how environmental factors can influence the expression of genes without changing the underlying DNA sequence — like flipping switches (nurture) on genes you were already born with (nature).

2. The Evolutionary Lens

The _____ studies the evolution of behavior and the mind using principles of natural selection. It relies heavily on _____ — Charles Darwin's principle that traits leading to increased reproduction and survival are more likely to be passed down to future generations.

_____ is the controversial, unethical practice or advocacy of improving the human species by selectively mating people with specific desirable hereditary traits — a dark chapter where evolutionary ideas were twisted to justify discrimination.

3. The Twin Playbook: How We Study This

_____ compare identical and fraternal twins to examine the relative influence of genetics and environment on behavior and traits.

- _____: identical twins formed when a single fertilized egg splits into two embryos, sharing nearly all the same genetic material.
- _____: fraternal twins formed when two separate eggs are fertilized by two different sperm cells, sharing about 50% of their genetic material — just like standard siblings.

By comparing twins raised together versus those raised apart, psychologists calculate _____ — a measure of how much differences in a trait among individuals in a population are due to genetic variation.

- _____: examine how often a trait or disorder shows up among relatives with varying degrees of genetic relatedness.
- _____: compare adopted children to both their biological relatives (shared genes, different environment) and adoptive relatives (shared environment, different genes).

Apply It: Which Research Method?

Using the word bank above, identify which research approach or twin type each scenario describes.

#	Scenario	Term
1	A pair of twins share about 50% of their genes, just like typical siblings, because they came from two separately fertilized eggs.	
2	Researchers compare rates of depression among siblings, cousins, and more distant relatives in the same extended family.	
3	Researchers track a child raised by adoptive parents, comparing her personality to both her biological mother and her adoptive mother.	

4. Don't Trip Up! (Common Misconceptions)

If a trait has a heritability of 60%, this does NOT mean 60% of one person's trait comes from genes. It means that within a specific _____, 60% of the variation (or differences) in that trait can be explained by genetic factors. Heritability applies to differences within a group, never to one individual.

"Mono" means _____ (one egg that split = identical twins). "Di" means _____ (two separate eggs = fraternal twins).

What's on the exam? Specific biological mechanics (like drawing chromosomes or explaining dominant vs. recessive alleles) are beyond the scope of the AP exam — focus on the relationship and interaction between nature and nurture instead.

5. Quick Check: Review Questions

Circle the best answer for each question below.

1. A student inherits genes that make her more likely to develop excellent musical ability, but because her family never exposes her to instruments or lessons, this potential never fully develops. This scenario best illustrates the concept of a:
 - A. Heritability Estimate
 - B. Genetic Predisposition
 - C. Family Study
 - D. Twin Study
2. Researchers find that severe childhood stress can cause certain genes related to mood regulation to become "switched off," even though the underlying DNA sequence itself never changes. This process is best described as:
 - A. Natural Selection
 - B. Heritability
 - C. Eugenics
 - D. Epigenetics
3. An evolutionary psychologist explains that humans commonly crave high-calorie, high-energy foods because ancestors who sought out these calorie-dense foods were more likely to survive periods of famine and pass down their genes. This principle, first proposed by Charles Darwin, is called:
 - A. Eugenics
 - B. Heritability
 - C. Natural Selection
 - D. Epigenetics
4. Twin A and Twin B formed when a single fertilized egg split into two separate embryos, giving them nearly identical genetic material. What type of twins are Twin A and Twin B?
 - A. Monozygotic Twins
 - B. Adopted Siblings
 - C. Dizygotic Twins
 - D. Fraternal Twins



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>